

Oxalosis

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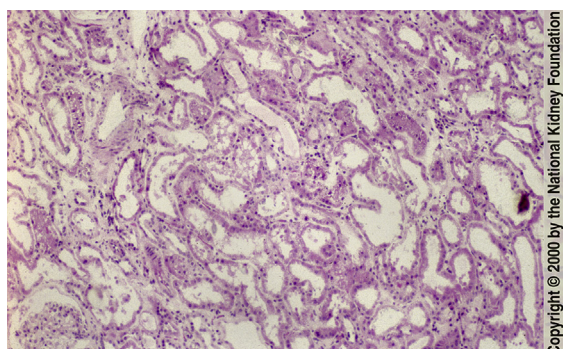


Fig 1. Oxalate crystals may be seen in relatively small amounts and in chronic renal disease. However, numerous oxalate crystals are characteristic of other conditions, such as primary oxalosis, jejunioileal bypass surgery, or ethylene glycol ingestion. By routine light microscopy, widespread tubular destruction is seen in this case of ethylene glycol ingestion. On closer examination, refractile crystals may be seen in some of the tubules. (Hematoxylin and eosin, original magnification $\times 100$).

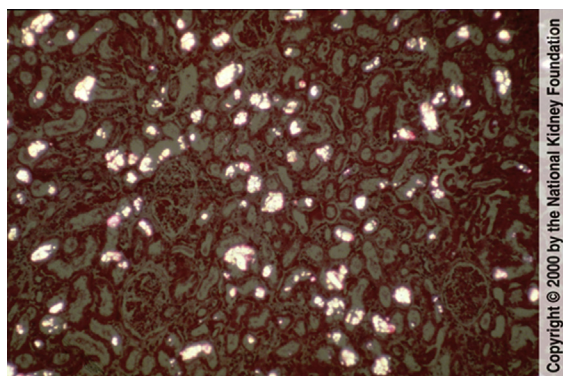


Fig 2. Crystals such as calcium oxalate are readily observed under polarized light microscopy. Here a polarized view is shown of the same case illustrated in the previous figure, with numerous calcium oxalate crystals characterized by birefringence and organizing in fan or sheaf-shaped arrays. There is associated tubular degeneration. (Polarized hematoxylin and eosin, original magnification $\times 100$).

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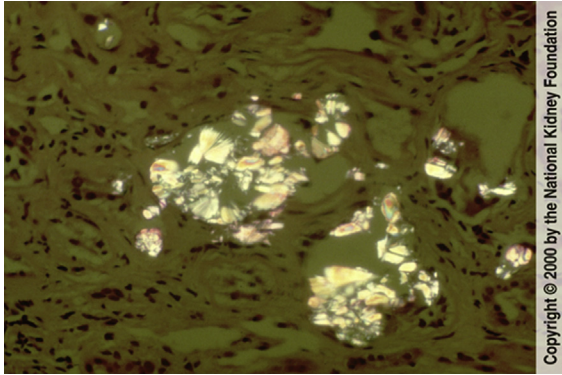


Fig 3. Calcium oxalate crystals are present within the tubular lumina in this polarized view of a renal biopsy from a patient with ethylene glycol intoxication. The calcium oxalate crystals show characteristic aggregation into fan or rosette-shape structures with occasional rhomboid individual crystals present. There is surrounding tubulointerstitial fibrosis and mononuclear cell infiltrate. (Polarized hematoxylin and eosin, original magnification $\times 200$).